

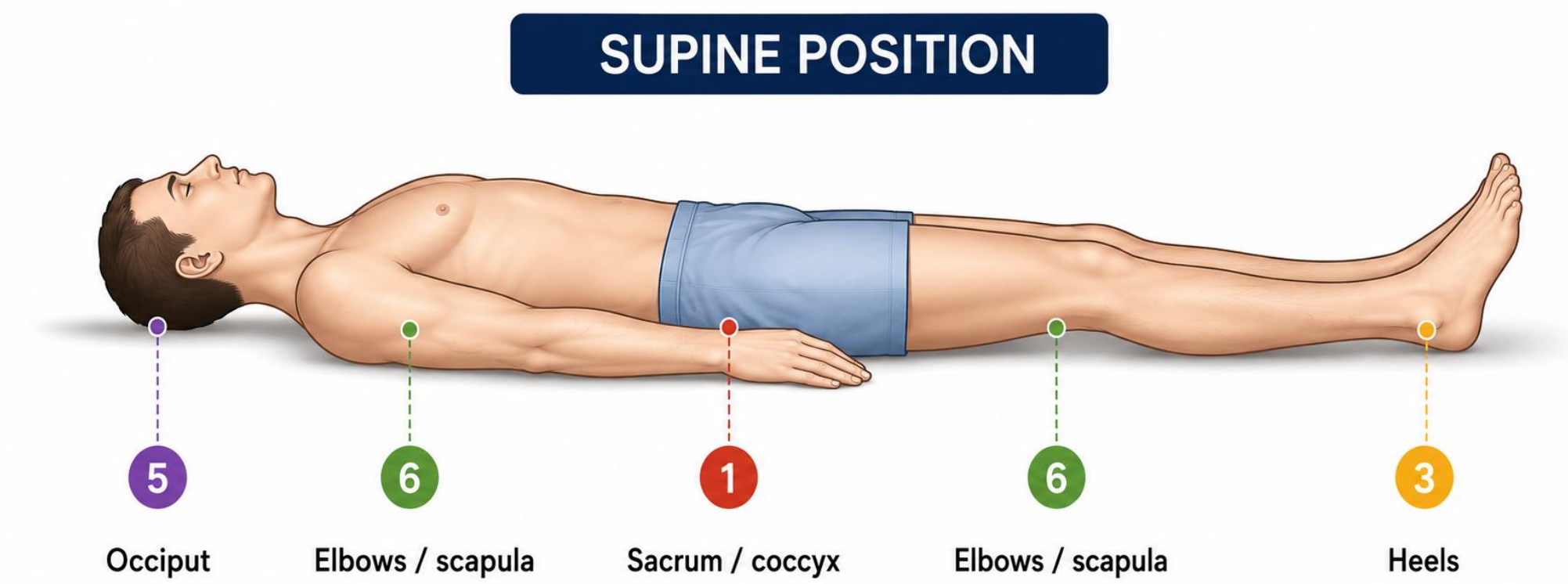
Smart Pressure and Temperature-based system for early detection of bedsores risk

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Motivation

- Bed sores, also known as pressure ulcers or decubitus ulcers, are localized injuries to the skin and underlying tissue that occur due to prolonged pressure, most commonly over bony prominences.
- Bedridden patients are at high risk of pressure ulcers due to prolonged pressure and heat buildup.
- Early tissue damage often occurs before visible skin changes.
- Current monitoring relies on manual checks and caregiver observation.
- There is a need for a low-cost, real-time bedsores risk detection system



State of the art

Alternating pressure air mattresses: KosmoCare Air Bed for Bedsores prevention (Mostly High Cost)

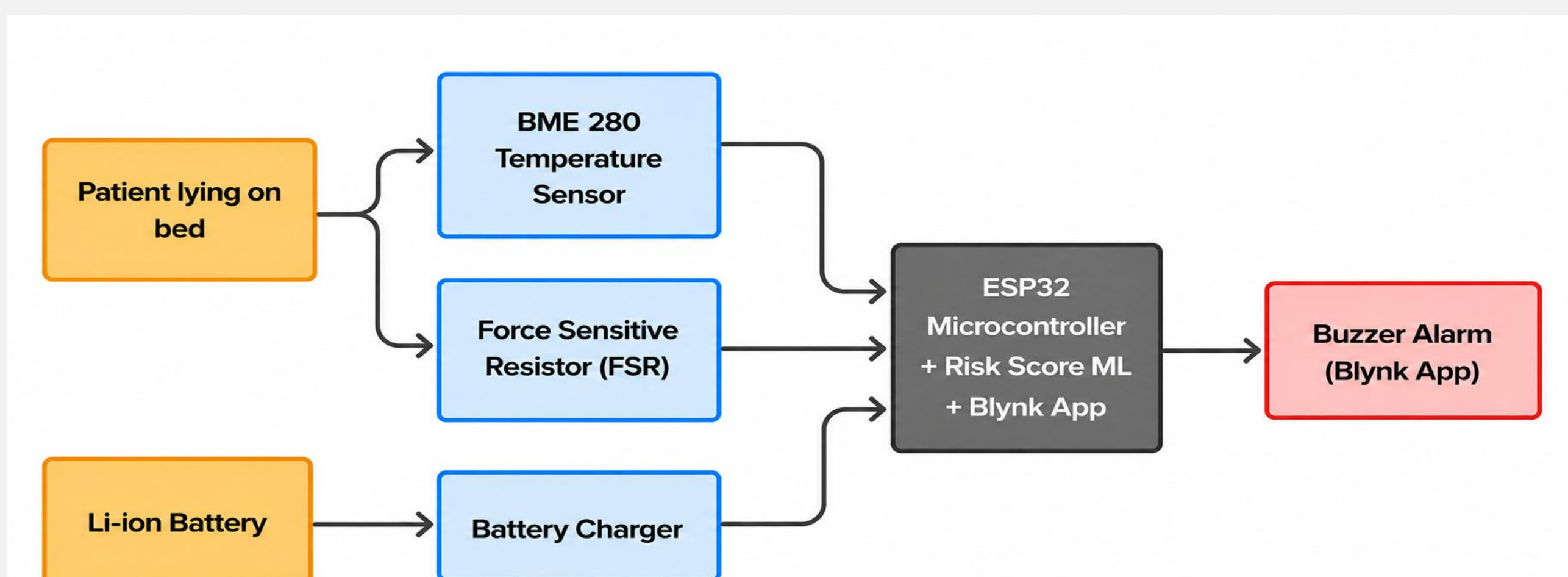
Smart sensor-based pressure monitoring:
SENSOMATT Pressure Monitoring Sensor Sheet, XSENSOR
Fore Site PT patient turn system (restricted to multi-speciality hospitals)

Research: IoT embedded prototypes exist

- In need of smart, low-cost, and real-time monitoring systems

Core technological idea

Long-duration pressure changes (dominant) detected by the sensor (Voltage) are converted into Force (N). If applied force is greater than threshold, the buzzer is triggered.



Implementation and Achievements

Arterial Capillary Closure Pressure: 32 mmHg (Baseline)

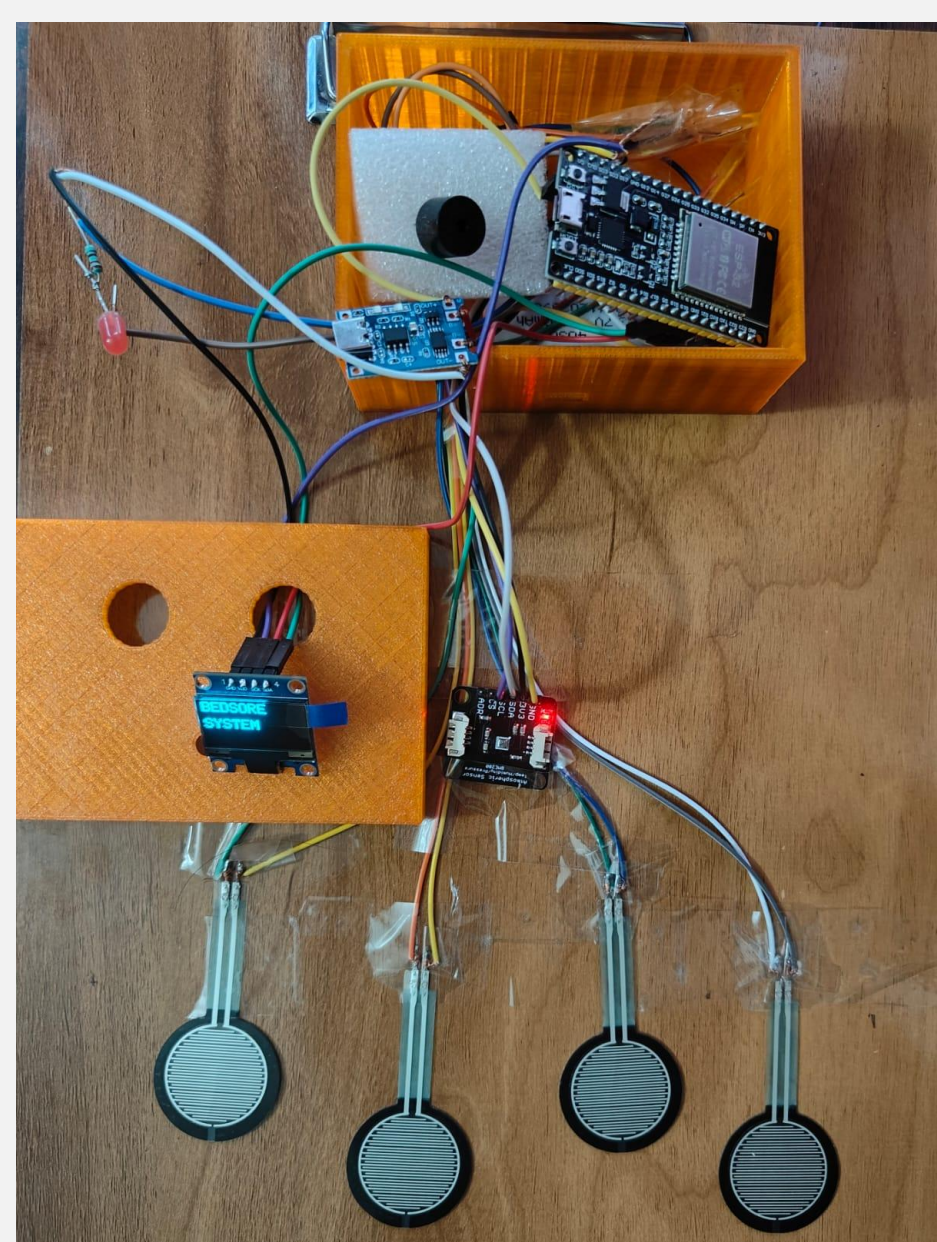
Average Indian Sacral Area: 100–105 cm²

Force = Pressure x Area (Pressure to force conversion)

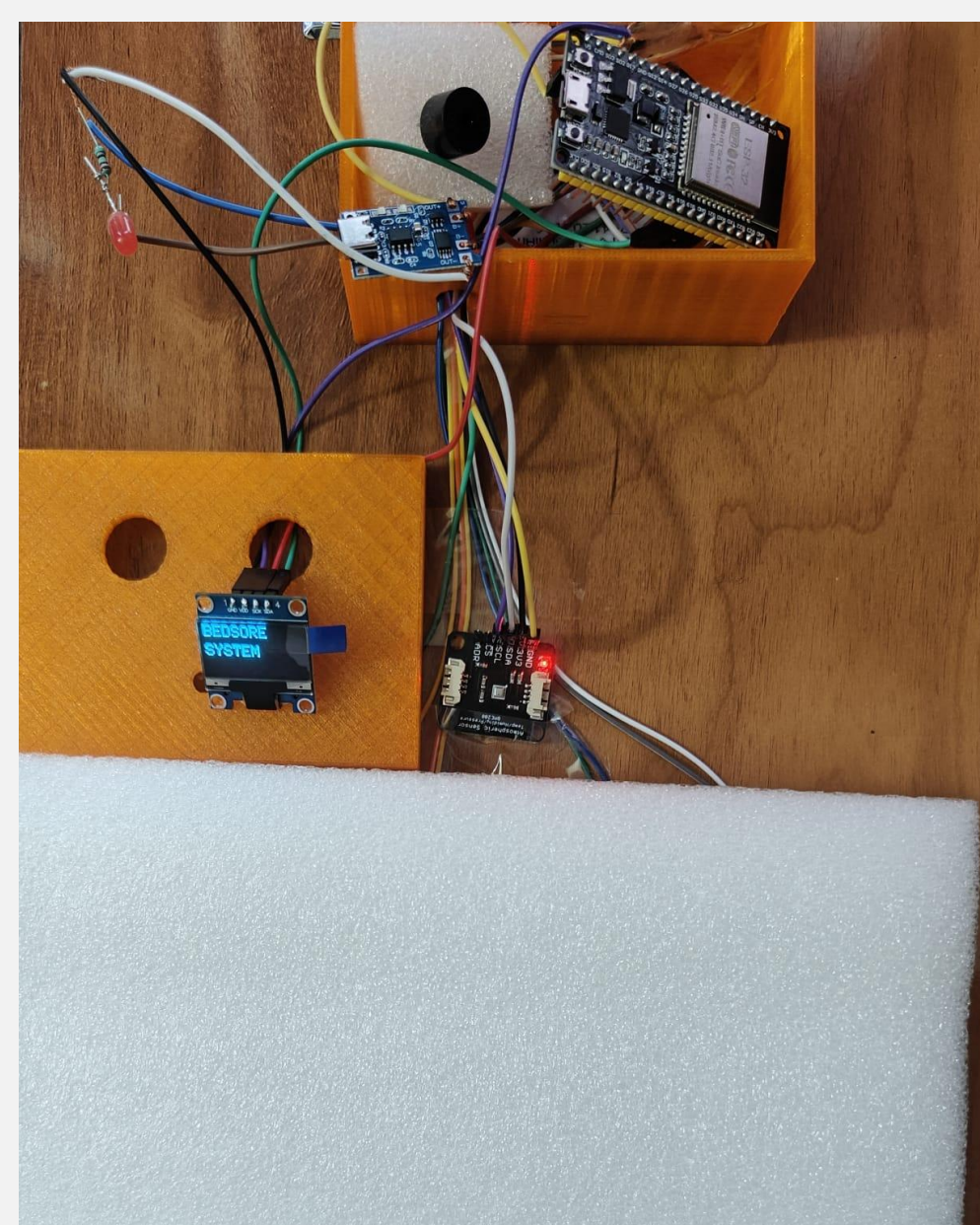
RiskScore = $(F \times 45) + (T \times 20) + (D \times 25) + (S \times 10)$

Critical Parameters:

Pressure (mmHg)	Local Skin Temperature Change (ΔT)	Risk Level	Interpretation
< 32	$\Delta T < 0.5^\circ\text{C}$	Low	Normal pressure & microclimate
32–40	$\Delta T 0.5\text{--}1.0^\circ\text{C}$	Moderate	Early tissue stress possible
40–60	$\Delta T 1.0\text{--}1.2^\circ\text{C}$	High	Reduced perfusion risk
> 60 or prolonged >32	$\Delta T > 1.2^\circ\text{C}$	Critical	High bedsores formation risk



Final Prototype



Blynk Interface

Reflection & Conclusion

- Development of skills in hardware prototyping, embedded programming, and system integration
- Successful design concept of a low-cost bedsores risk detection system
- Sensor placement and threshold calibration and risk model prediction may need improvement
- The system has potential for early bedsores risk detection in bedridden patients

Outlook

- Improvement of wearable cushion design for patient comfort
- Long-term reliability training and validation of the monitoring system

References

- [1] NPUAP, Pressure Ulcer Prevention Guidelines, 2019
- [2] Minter et al., Pressure Ulcer Monitoring Platform (PUMP), Journal of Wound Care, 2019
- [3] A. Memari et al./ Transactions on Machine Intelligence 3(2) (2020) 101-110